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Project 3 SSD Performance Profiling

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Introduction

This project examines how access pattern, block size, read/write mix, queue depth and parallelism affect SSD throughput and latency. fio measurements cover QD 1 baselines, throughput–latency curves, completion-latency tails, working-set changes and a sustained write window. The results show that additional concurrency helps until throughput gains diminish, while write traffic produces substantial stalls and long tails. A warmed buffered-read control also shows how caching can produce a misleading SSD performance claim.

System Configuration

Item	Configuration and evidence
Processor	AMD Ryzen 5 PRO 7530U with Radeon Graphics; 6 cores, 12 logical processors; advertised base 2.00 GHz.
Host	Lenovo 21H6S04400; Windows 11 Education, build 26200. Host reports 15,886,163,968 bytes of physical memory.
SSD	SKHynix_HFS512GEJ4X113N; firmware 51790C30. PhysicalDisks reports NVMe, SSD, Healthy and 512,110,190,592 bytes. Win32_DiskDrive reports 512,105,932,800 bytes and a SCSI interface label.
Guest and tools	Ubuntu distribution under WSL2; WSL 3.0.3.0; kernel 6.18.54.1-microsoft-standard-WSL2; x86_64; Python 3.10.12; fio 3.28.
Target and filesystem	/home/yilmat2/ssdbench-data/workload.bin; 17,179,869,184 bytes (16 GiB); ext4 on guest /dev/sdd. This is a managed regular file inside the WSL virtual disk.
Memory and space	Guest MemTotal 7,688,994,816 bytes; 2 GiB swap, unused at start. Guest free space 225,302,908,928 bytes. Earlier Windows snapshot: C: NTFS, 101,065,531,392 bytes free.
Power and temperature	Windows snapshot reports Balanced power mode and SSD temperature 40 °C. Its TemperatureMax field is 86 °C; it is not a measured peak during this experiment. AC state and cooling were not recorded.

Host metadata identifies NVMe despite the compatibility SCSI label. The exact Windows path of the distribution's VHDX and its volume-to-SSD mapping were not recorded. Results therefore describe the complete WSL storage path; they do not isolate bare-device performance. [2]

Run Procedure

The full run contains 105 configurations with three successful measured trials each, giving 315 retained runs. The 97 static configurations comprise four baselines, 24 block-size cases, eight mixes, 36 queue-depth cases, 16 worker-count cases, six working-set cases, two cache controls and one sustained-write case. Eight adaptive tail configurations add 24 runs. The journal spans October 23, 2026, 9:22–11:48 p.m. EDT (October 24, 01:22–03:48 UTC), including the interruption and resume.

The target was initialized by one full sequential write over all 16 GiB before measurement. The saved ownership record reports successful initialization and records its raw-result hash; initialization files themselves were outside the supplied results archive. The runner checks ownership, file type, size, inode and device, rejects unsafe targets and locks the directory. It never selects a raw partition. Aligned block sizes and disjoint worker offsets keep I/O inside the managed file. Initialization is not a full-drive steady-state precondition.

Most cases use libaio, direct=1, one worker, 15 measured seconds after a three-second ramp and a two-second cooldown. Tail cases run for 45 measured seconds; the sustained write runs for 120 seconds without a ramp. Each trial randomizes static-case order with seed 7530 and records per-case seeds. Data buffers are refilled and scrambled with buffer_compress_percentage=0, targeting incompressible data. The two cache controls use psync at QD 1 and the same warmed 256 MiB range; only direct changes.

IOPS counts requests per second. MiB/s uses 2^{20} bytes per MiB; KiB and GiB are also binary units. Total latency includes submission and completion portions; percentiles describe completion latency. [1] Means and error bars summarize three trial statistics using sample SD, not confidence intervals or individual-I/O SD. Percentiles are obtained from JSON+ histogram bins; worker histograms are merged within a run. Figure 11 instead pools histogram counts across trials. Calculations use full CSV precision and displayed results round to four decimal places; no values are read from plot coordinates.

Code Structure and Execution Flow

Component	Responsibility
scripts/ssdbench.py and cli.py	Expose inspect, init, plan, run, analyze, plot and guarded cleanup commands.
plan.py and safety.py	Build the experiment matrix and fio jobs; validate the owned file, bounds, worker regions and lock.
runner.py and environment.py	Execute fio, record actual order and seeds, collect system snapshots, hash raw outputs and resume verified completed trials.
analysis.py and plotting.py	Parse per-worker JSON+, summarize repetitions, select knee candidates, evaluate Little’s Law and generate 13 figures.
collect_host.ps1 and build_report.py	Collect Windows hardware metadata and provide the initial editable report framework.

Hypothesis

Larger blocks are expected to increase byte throughput while reducing request rate. Additional queue depth should first expose more parallel work, then raise response time with little further throughput. Mixing reads and writes may create interference rather than interpolate between pure workloads. Tail latency is expected to be more sensitive than the mean to intermittent stalls. Warm buffered reads are expected to measure cache service rather than establish SSD capability.

Experiments and Results

QD 1 Baselines

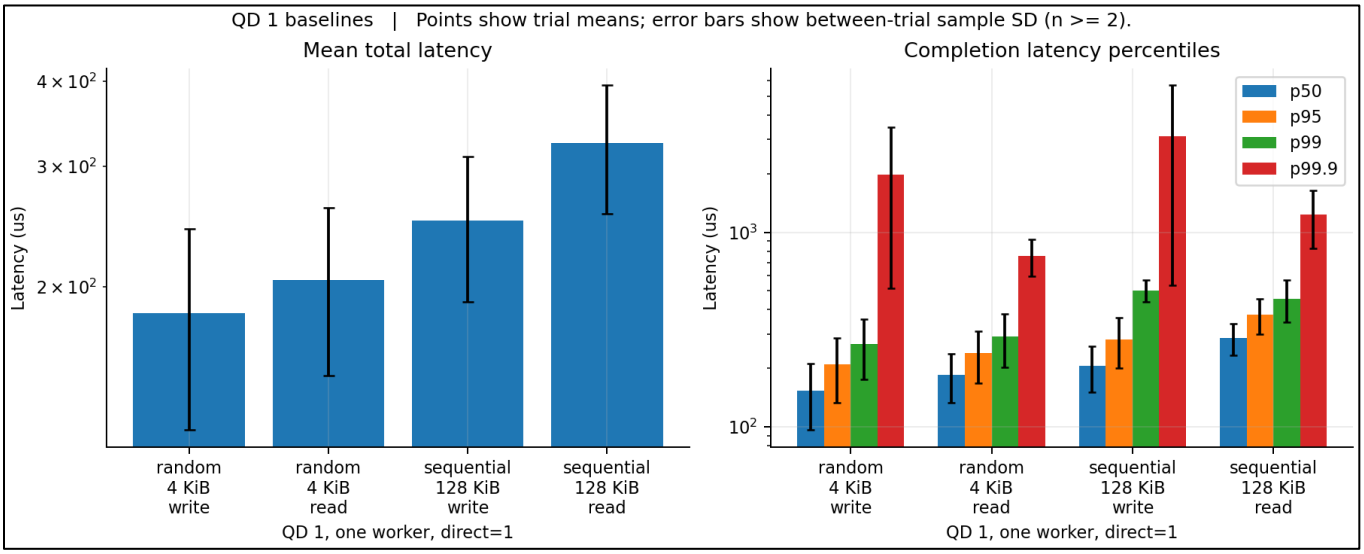


Figure 1. QD 1 total-latency means and completion percentiles. Each point summarizes three trials; bars show between-trial sample SD.

Workload	Mean total µs mean ± SD	p50 µs	p95 µs	p99 µs	p99.9 µs
random 4 KiB read	204.6076 ± 56.4475	184.6613	238.9333	292.1813	757.7600
random 4 KiB write	183.0809 ± 59.6910	153.9413	208.8960	266.9227	1983.8293
sequential 128 KiB read	324.9582 ± 69.5248	285.6960	376.1493	456.0213	1235.6267
sequential 128 KiB write	250.1879 ± 60.1666	205.4827	281.2587	503.8080	3112.9600

QD 1 is the project's low-queue baseline: one benchmark request is outstanding at a time. It still includes Linux, virtualization, filesystem, host and SSD service. Random 4 KiB reads average 204.6076 µs total latency, versus 183.0809 µs for writes. Sequential 128 KiB reads average 324.9582 µs, versus 250.1879 µs for writes. Different block sizes prevent interpreting the random/sequential comparison as an isolated access-order penalty.

Write completion tails are more extended than their medians suggest. Sequential writes have a mean p50 of 205.4827 µs and mean p99.9 of 3112.9600 µs; the p99.9 between-trial SD is 2580.4800 µs. The corresponding random-write p99.9 is 1983.8293 ± 1470.3347 µs. A single central-latency number would hide this variability. Writes completing faster at QD 1 do not establish faster persistent NAND programming: the experiment does not force a durability barrier after every write.

Workload	IOPS mean ± SD	Completion samples per trial
random read	5067.0400 ± 1212.2608	55,132–88,389
random write	5737.7064 ± 1613.7523	58,866–105,235
sequential read	3145.3014 ± 612.3440	36,881–54,521
sequential write	3765.5045 ± 842.5309	42,278–66,477

Baseline completion populations range from 36,881 to 105,235 samples per trial. Most baseline trials therefore have fewer than 100 expected observations beyond p99.9. These tail estimates remain useful descriptions of the recorded intervals, but their stability is limited by both population size and only three repetitions.

Block Size and Access Pattern

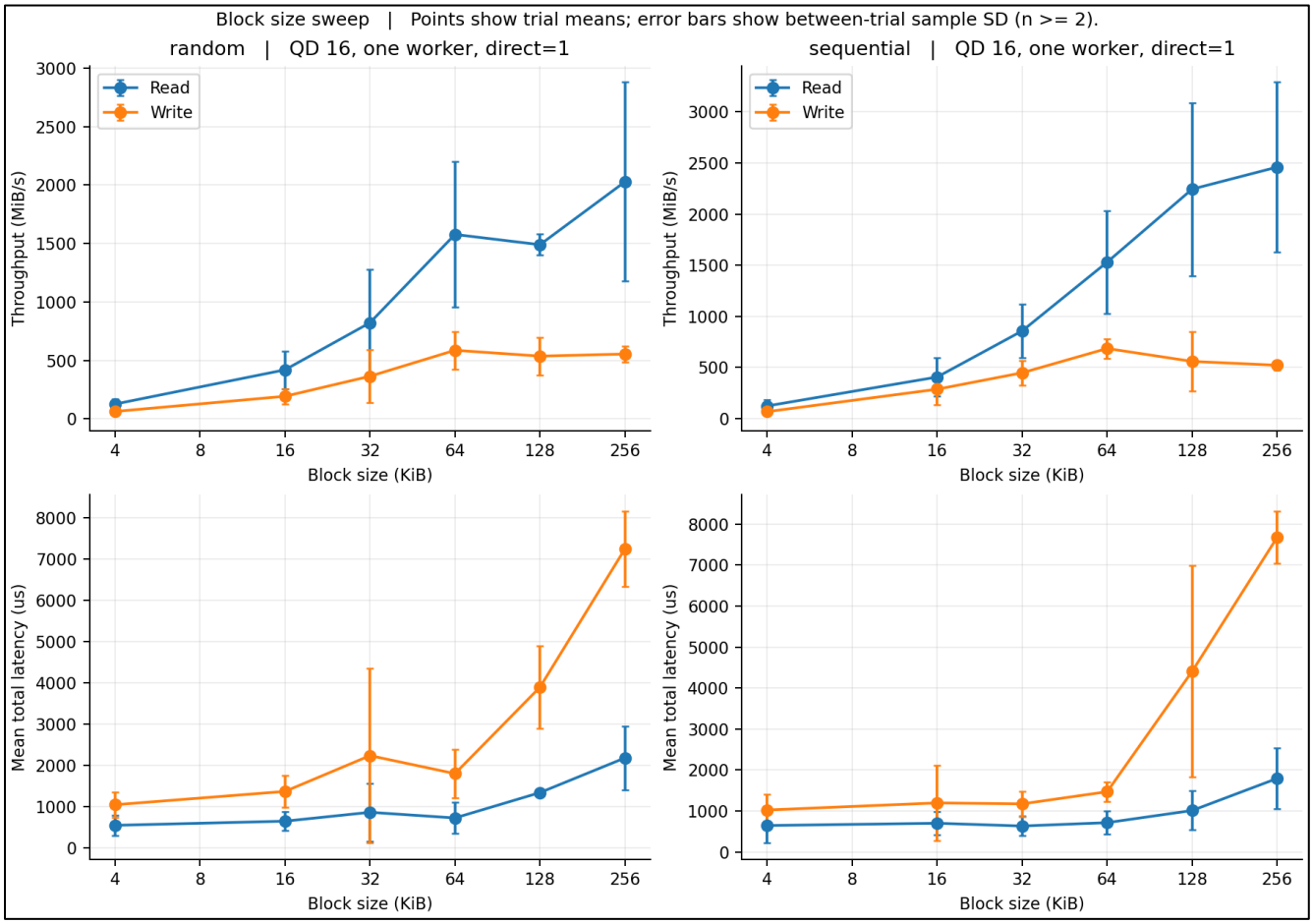


Figure 2. Block-size sweep at QD 16, one worker and direct I/O. Throughput and total latency come from the same runs.

At fixed QD 16, sequential read throughput rises from 122.7365 ± 58.5788 MiB/s at 4 KiB to 2459.5672 ± 830.6421 MiB/s at 256 KiB. Random reads rise from 126.5805 ± 45.3770 to 2026.9113 ± 851.2127 MiB/s. Larger requests move more bytes per completion, even as the rate of completed requests declines. The large SDs caution against treating neighboring means as cleanly separated limits.

Write throughput peaks at 64 KiB within this block sweep: 586.5801 ± 160.6704 MiB/s randomly and 684.9182 ± 95.1850 MiB/s sequentially. At 256 KiB, the corresponding means are 554.5495 and 520.9904 MiB/s. Mean total write latency rises from 1050.8294 to 7245.7026 µs randomly and from 1022.1552 to 7675.7258 µs sequentially between 4 and 256 KiB. At this fixed concurrency, making requests larger eventually increases waiting and service time without improving write throughput.

Request Rate and Read Write Mix

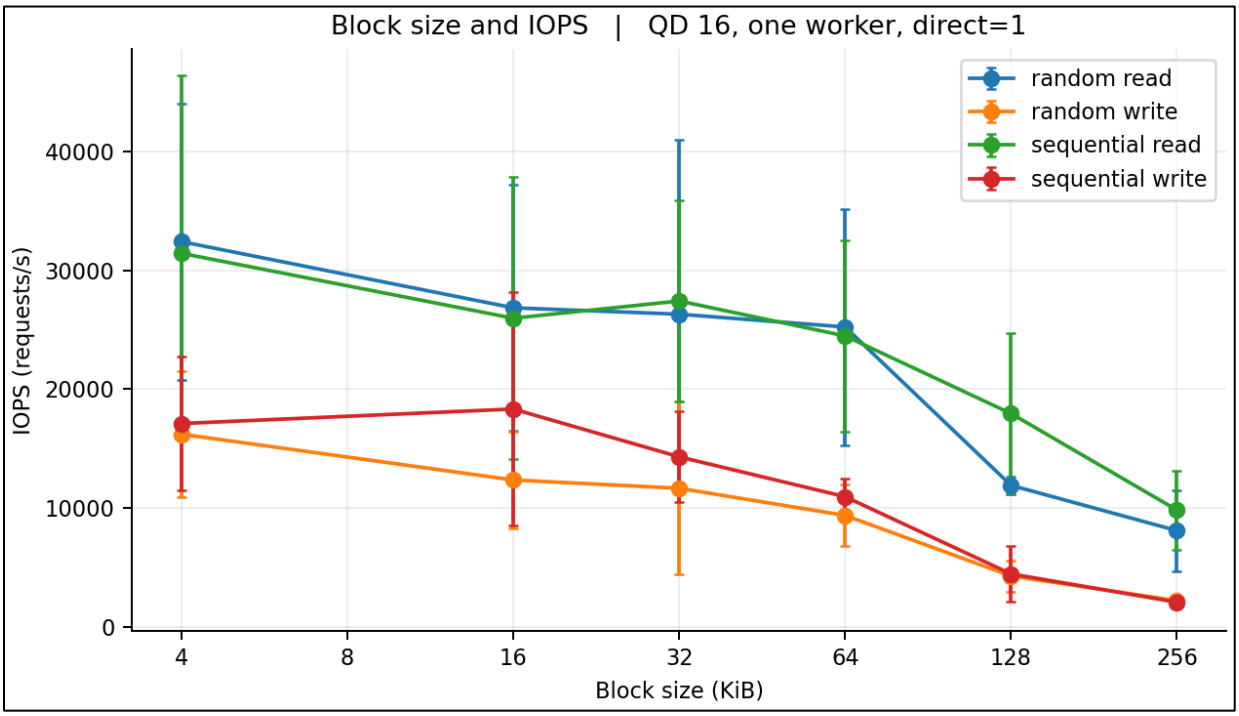


Figure 3. IOPS for the block-size sweep. Block size changes request rate and byte throughput in different ways.

Random-read IOPS fall from 32403.5953 at 4 KiB to 8106.6453 at 256 KiB; sequential reads fall from 31419.5029 to 9837.2244. For a fixed block size, bandwidth equals IOPS × bytes per request. A larger byte rate can therefore coexist with fewer requests per second; it does not imply lower request latency.

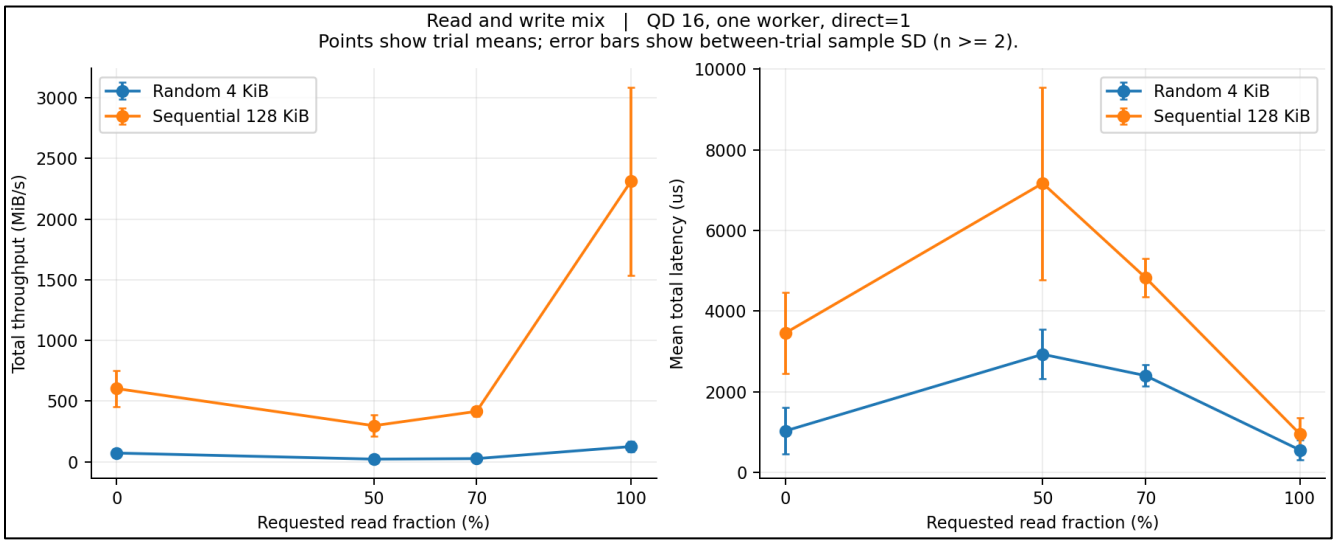


Figure 4. Requested read fraction at QD 16 and one worker. Both panels use aggregate read-plus-write statistics.

At 100/0, 70/30, 50/50 and 0/100 read/write ratios, random 4 KiB throughput is 125.5592, 26.3027, 21.9704 and 71.9806 MiB/s. Sequential 128 KiB throughput is 2311.3977, 415.7153, 297.6269 and 603.5030 MiB/s. Both mixed workloads are slower than either pure endpoint, so a weighted average of pure rates would be misleading. The mean achieved read fractions are 69.9949% and 50.1542% randomly, and 69.8802% and 50.0330% sequentially.

For sequential traffic, mean total latency is 952.6597 µs for pure reads, 4828.4017 µs at 70/30 and 7168.6532 µs at 50/50. Read/write interference in the measured storage path is consistent with this loss, but these measurements do not separate controller scheduling, flash management and host effects. The pure endpoints here are separately repeated mix-family cases, rather than reused block-sweep observations.

Throughput Latency Knee

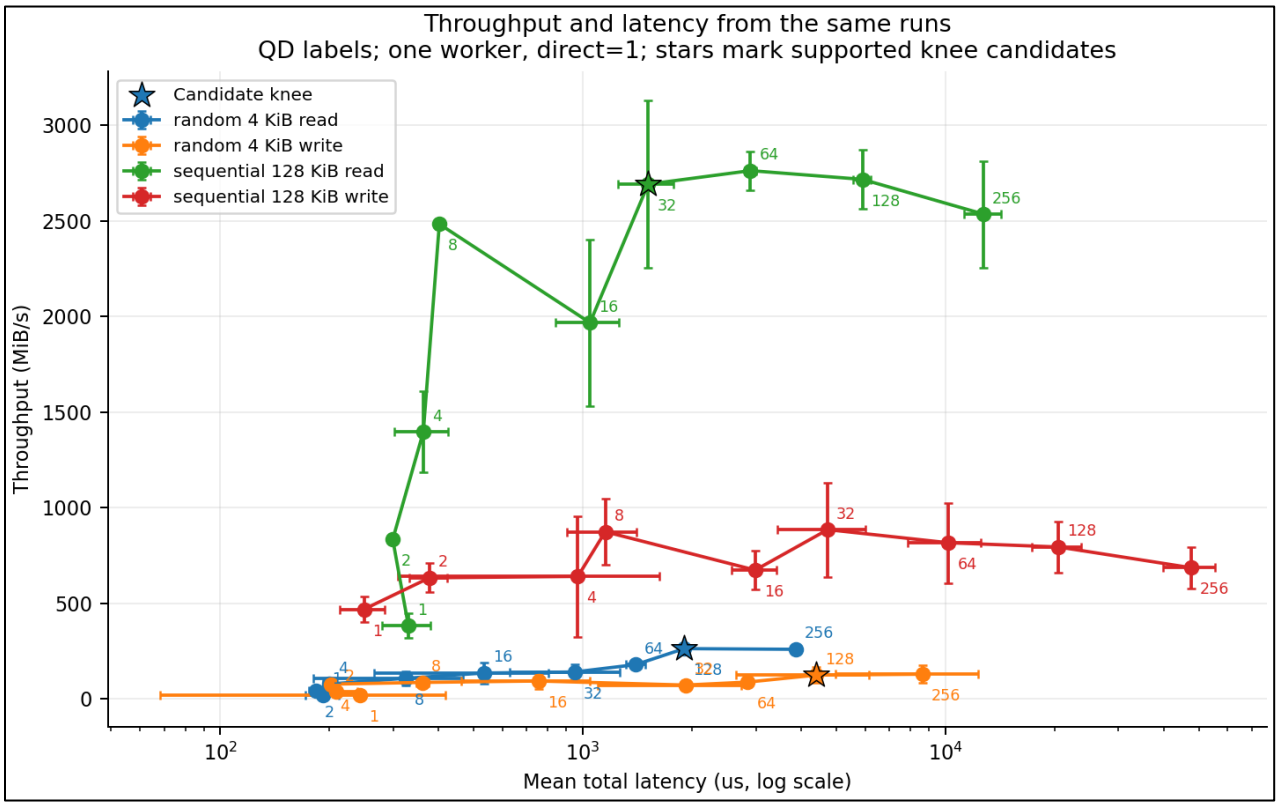


Figure 5. Throughput versus total latency for the queue sweep, QD 1–256. Stars mark supported operational knee candidates.

The declared knee rule uses trial medians: both adjacent depths must reach at least 90% of that workload's peak median throughput, the higher depth must gain no more than 10% throughput, and median total latency must rise by at least 25%. The lower depth is selected. Figure 5 displays trial means, so its points need not reproduce the median-based decision visually. A fallback at the highest depth is explicitly not a demonstrated knee.

Workload	Candidate QD	Median MiB/s lower → next	Median total µs lower → next
Random 4 KiB read	128	261.0032 → 260.3129	1915.1289 → 3841.2246
Random 4 KiB write	128	146.7460 → 136.6042	3405.8489 → 7319.8567
Sequential 128 KiB read	32	2804.0559 → 2811.1711	1425.5831 → 2844.9687
Sequential 128 KiB write	Not established	QD 256 is a fallback	No adjacent pair passed

Read throughput can flatten while latency roughly doubles: sequential reads gain little between QD 32 and 64, and random reads lose a small amount between QD 128 and 256. This supports avoiding extra concurrency after its throughput benefit fades. The write curve is irregular and variable; an unsupported sequential-write knee should not be inferred from a local maximum.

Queue Depth and Little's Law

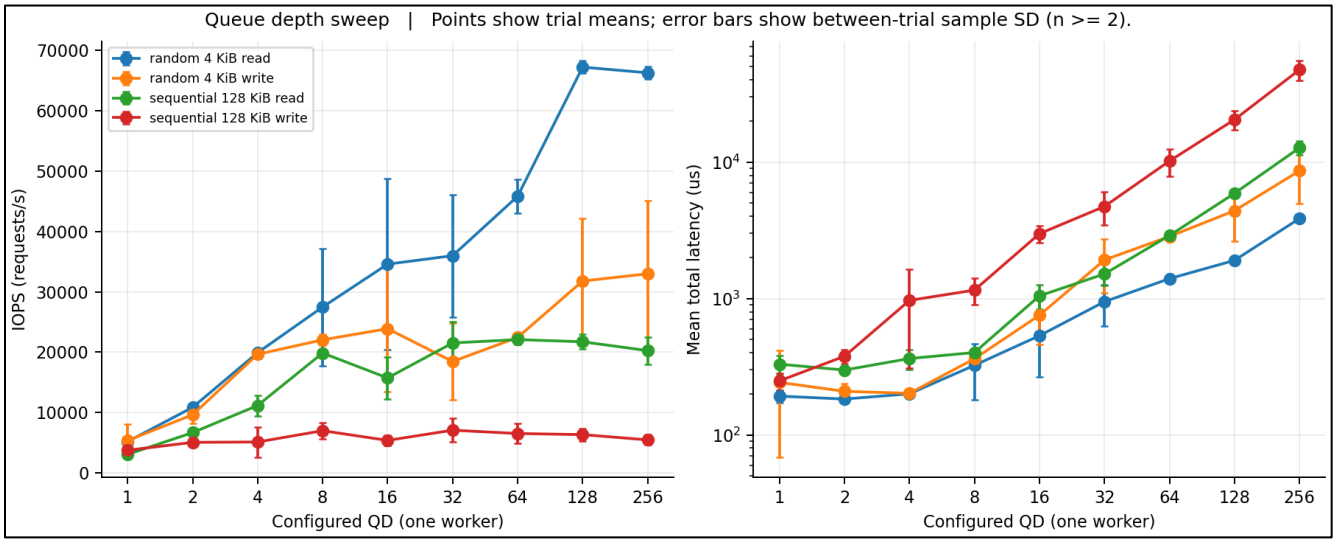


Figure 6. IOPS and mean total latency versus configured queue depth. Error bars show sample SD across three trials.

Random-read mean IOPS rise from 5188.8985 at QD 1 to 67244.4119 at QD 128, then fall to 66306.0295 at QD 256. Mean total latency rises from 192.3900 to 1903.0309 and 3860.2085 µs. Increased concurrency exposes more work to the storage stack, then mainly adds queued time. Controller and NAND parallelism are possible contributors, but the virtual path, generator and host also affect this curve; the data do not identify which resource sets its ceiling.

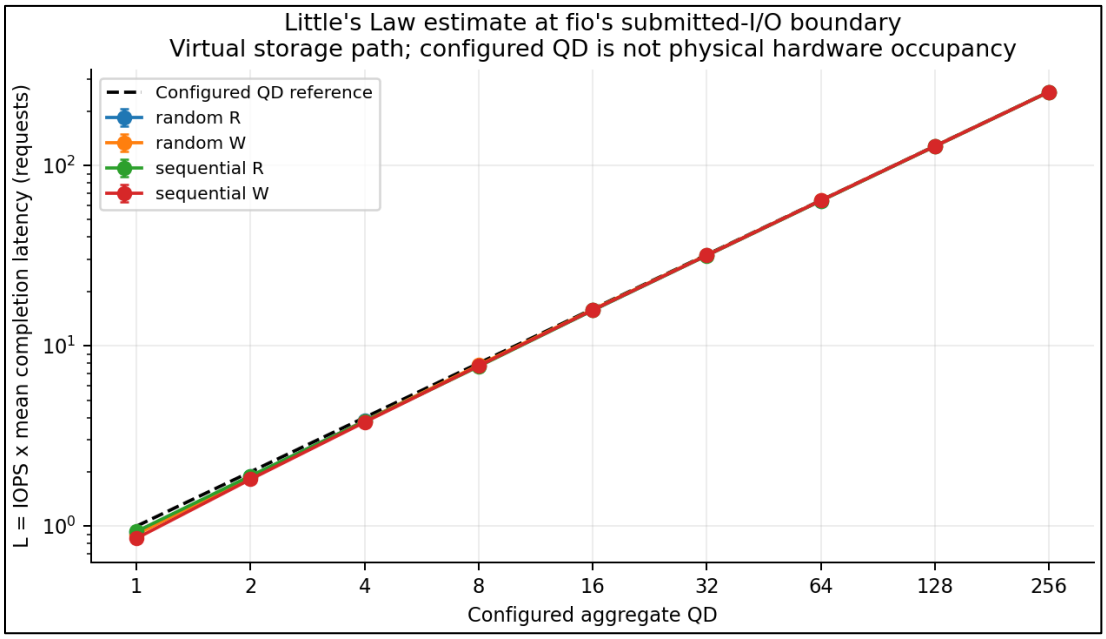


Figure 7. IOPS × mean completion latency estimates outstanding requests at fio's submitted-I/O boundary, not physical SSD queue occupancy.

Little's Law relates average outstanding work L, completion rate λ and response time W for the same population: L = λW. In trial 1 of QD 128 random reads, λ = 68429.076123 IOPS and W = 1863.61835708 µs, giving L = 127.5257 requests. Using total latency instead gives 127.9480. This is consistent with a mostly filled configured queue at fio's boundary; it does not measure NVMe controller entries or NAND operations.

The three-trial mean completion-boundary estimate is 127.5209 at random-read QD 128, and 31.6436 at sequential-read QD 32. These are averages of per-trial products, not products of independently averaged rate and latency. The small ramp-boundary sample discrepancy limits exact population matching. fio's highest depth bucket is ≥64, so its bucket data cannot distinguish exact occupancy at QD 128 and 256. [1]

Parallel Workers and Working Set

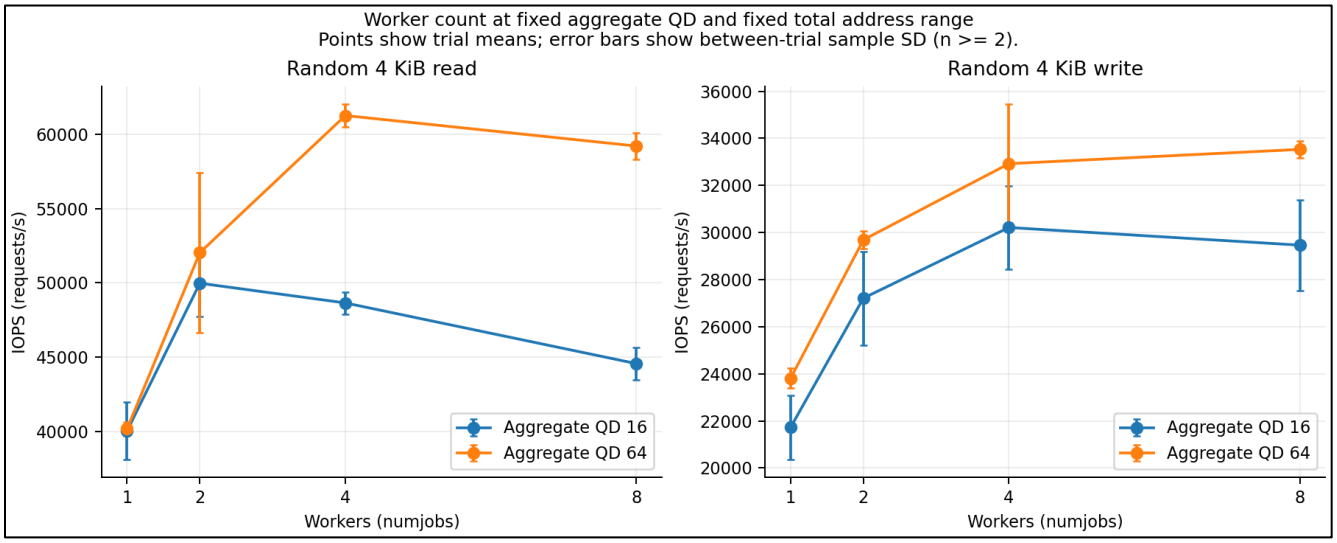


Figure 8. Random 4 KiB worker sweep at aggregate QD 16 or 64. Total address range remains 16 GiB; worker regions are disjoint.

At aggregate QD 16, random reads increase from 40017.2211 IOPS with one worker to 49977.3793 with two, then fall to 44555.6741 with eight. At aggregate QD 64, four workers reach 61284.5245 ± 754.4860 IOPS, versus 40154.5127 ± 459.8329 for one. Random writes at aggregate QD 64 rise from 23814.6049 with one worker to 33530.1061 with eight. Holding aggregate depth fixed shows that work distribution and submission concurrency matter separately from the number of outstanding requests.

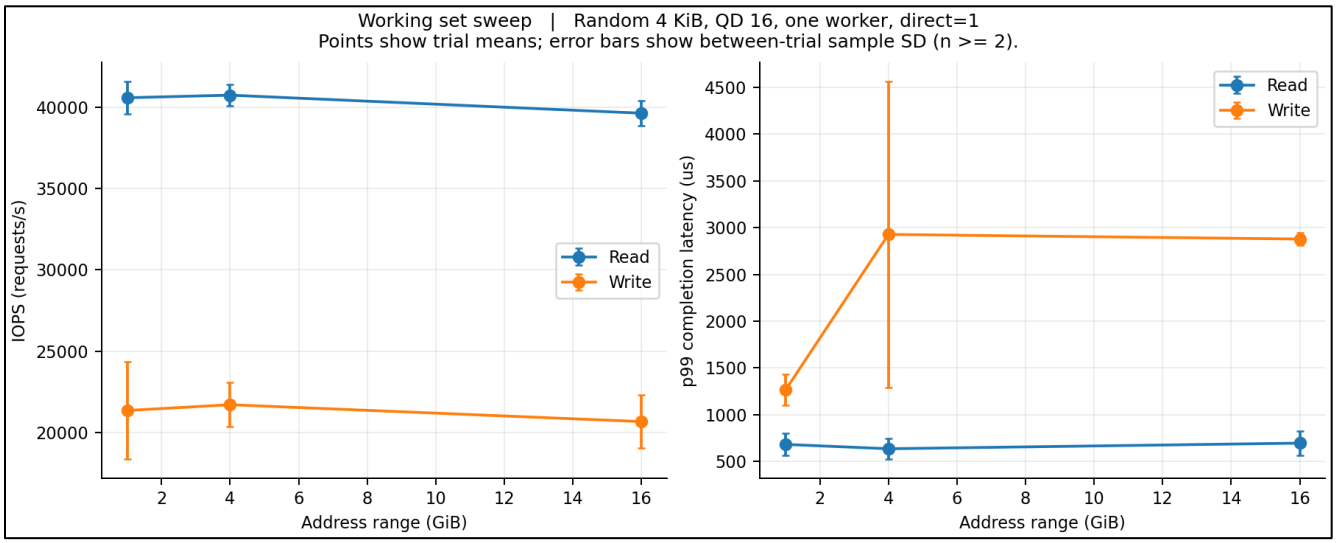


Figure 9. Working-set sweep at random 4 KiB, QD 16, one worker and direct I/O. Address ranges are 1, 4 and 16 GiB.

Read IOPS are 40567.2066, 40725.4183 and 39618.4032 across 1, 4 and 16 GiB. Write IOPS are 21372.2647, 21726.1960 and 20691.3984. The small changes do not establish a strong working-set capacity boundary. Read p99 is 681.3013, 634.8800 and 694.9547 µs; write p99 is 1264.2987, 2927.2747 and 2878.1227 µs. Large write-tail variability and only three ranges limit interpretation. Direct I/O reduces guest page-cache effects, while host/device caches and flash mapping remain possible influences.

Tail Latency at Selected Depths

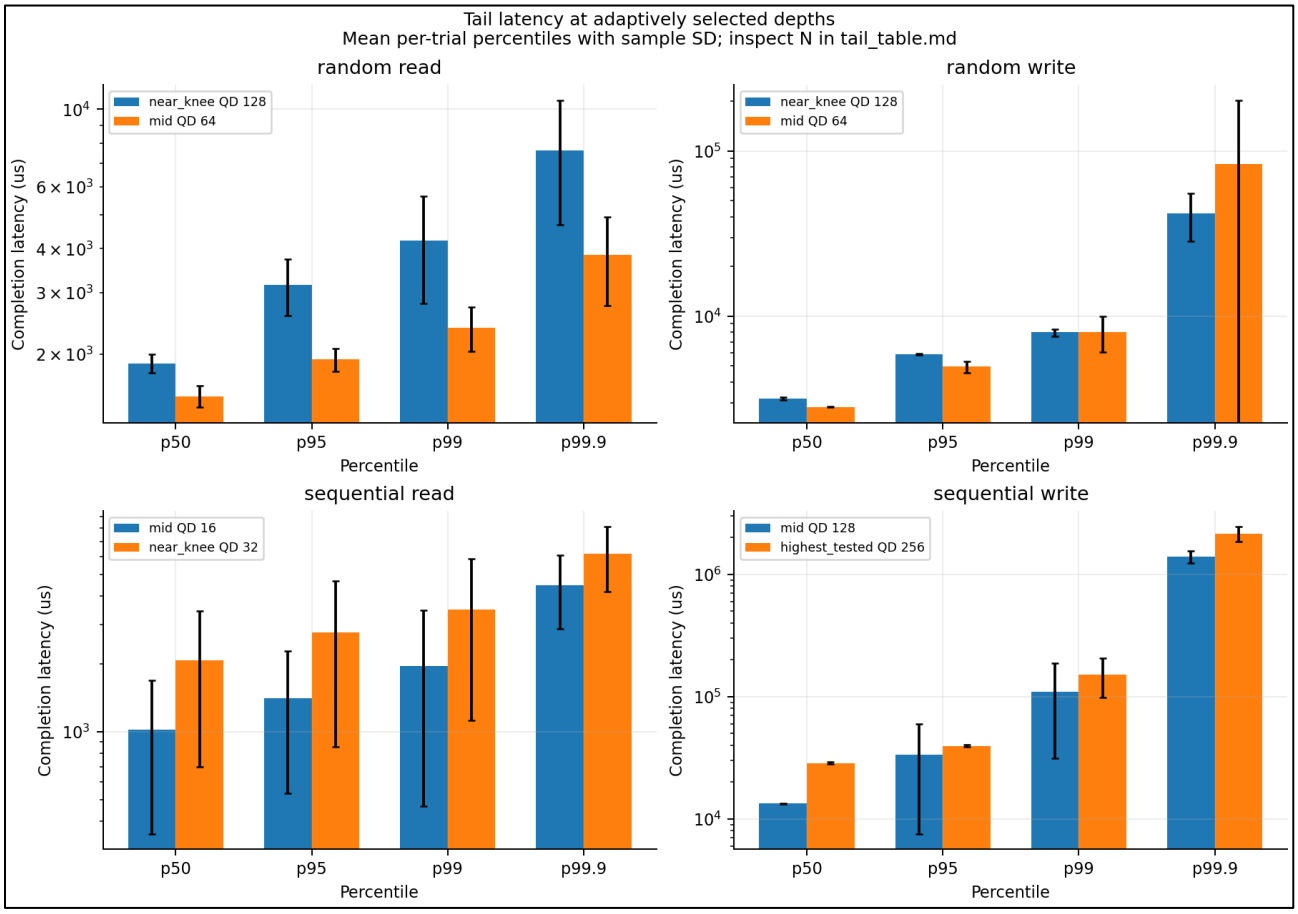


Figure 10. Mean per-trial completion percentiles with sample SD. Sequential-write QD 256 is the highest-tested fallback, not a supported knee.

Workload and QD	p50 ms	p95 ms	p99 ms	p99.9 ms
Rand R QD 64	1.5155	1.9306	2.3784	3.8393
Rand R QD 128	1.8814	3.1512	4.2161	7.6459
Rand W QD 64	2.8235	4.9480	8.0172	83.3181
Rand W QD 128	3.1730	5.8655	7.9845	41.8993
Seq R QD 16	1.0213	1.4036	1.9606	4.4783
Seq R QD 32	2.0698	2.7580	3.4816	6.1604
Seq W QD 128	13.3475	33.6418	109.2704	1400.8975
Seq W QD 256	28.5300	39.4090	151.5192	2153.0761

Random-read p99.9 rises from 3.8393 ms at QD 64 to 7.6459 ms at QD 128. Sequential-read p99.9 rises from 4.4783 to 6.1604 ms between QD 16 and 32. These longer trials are separate observations from the 15-second queue sweep; they are not pooled into its candidate selection.

Sequential-write p99.9 reaches 1400.8975 ± 160.0444 ms at QD 128 and 2153.0761 ± 306.4235 ms at QD 256, far above p50 values of 13.3475 and 28.5300 ms. Conversely, random-write p99.9 falls from 83.3181 to 41.8993 ms between the selected depths. Its QD 64 SD is 119.4705 ms, so this reversal does not support a general claim that deeper queues improve write tails. Intermittent events and run state strongly affect the observed extreme percentile.

Tail Populations and Survival Curves

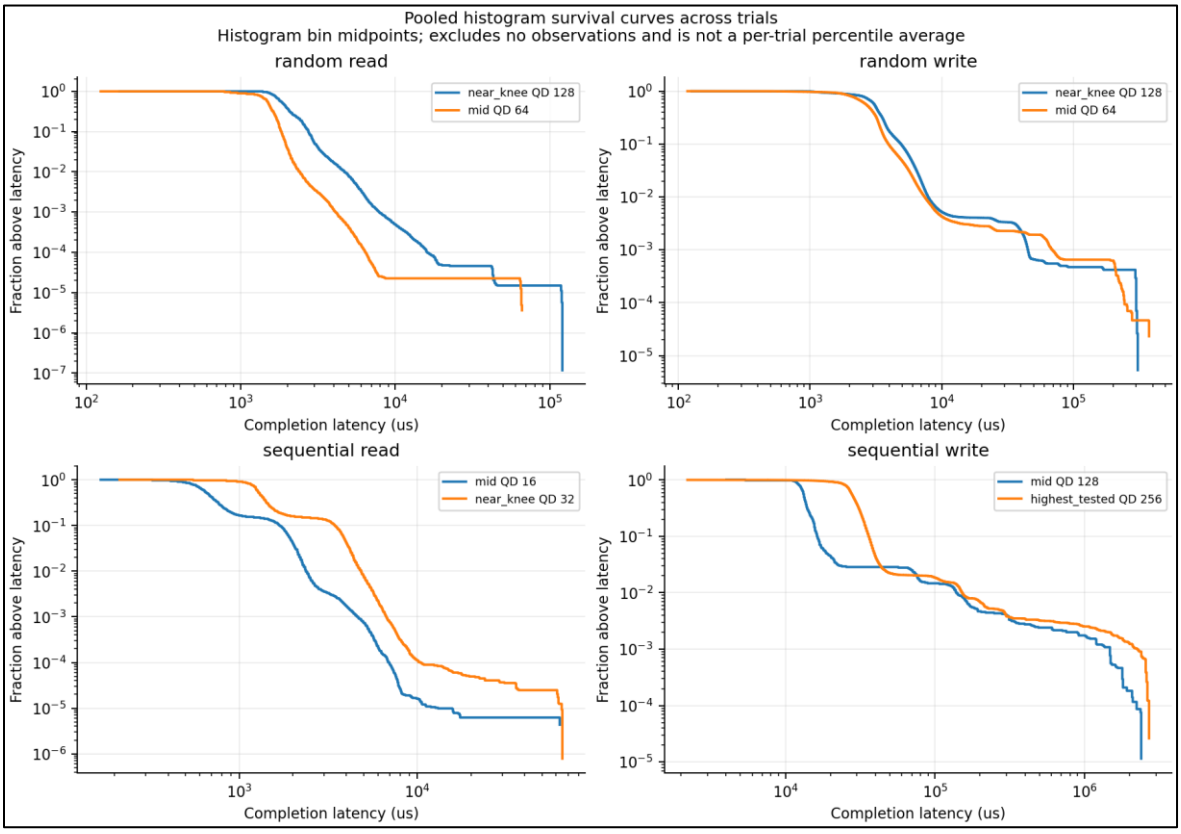


Figure 11. Survival fraction above each completion-latency bin midpoint. Counts are pooled across all three trials for each configuration.

The pooled survival curves preserve the long write tails that central tendency hides. Sequential writes retain a small population of completions lasting hundreds of milliseconds to seconds. Increasing read depth generally shifts more of the latency population to the right. Figure 11 weights each trial by its histogram population; Figure 10 and the preceding table give equal weight to each trial's percentile. A pooled percentile and the mean of per-trial percentiles are different statistics.

Tail configuration	Completion samples per trial
Random read QD 64	1,739,784–2,135,072
Random read QD 128	2,562,916–3,011,955
Random write QD 64	764,421–1,018,820
Random write QD 128	1,560,479–1,695,531
Sequential read QD 16	390,770–1,075,742
Sequential read QD 32	390,081–1,103,597
Sequential write QD 128	268,149–325,124
Sequential write QD 256	311,756–325,153

Every tail trial contains at least 268,149 completion samples, providing at least 268 expected observations above p99.9 under a stable distribution. This improves population support over the short baselines, but does not create 268 independent repetitions: temporally clustered stalls and changing device state can still dominate. Percentiles inherit histogram-bin resolution, and three trials provide only a limited estimate of run-to-run variation.

Sustained Writes and Device State

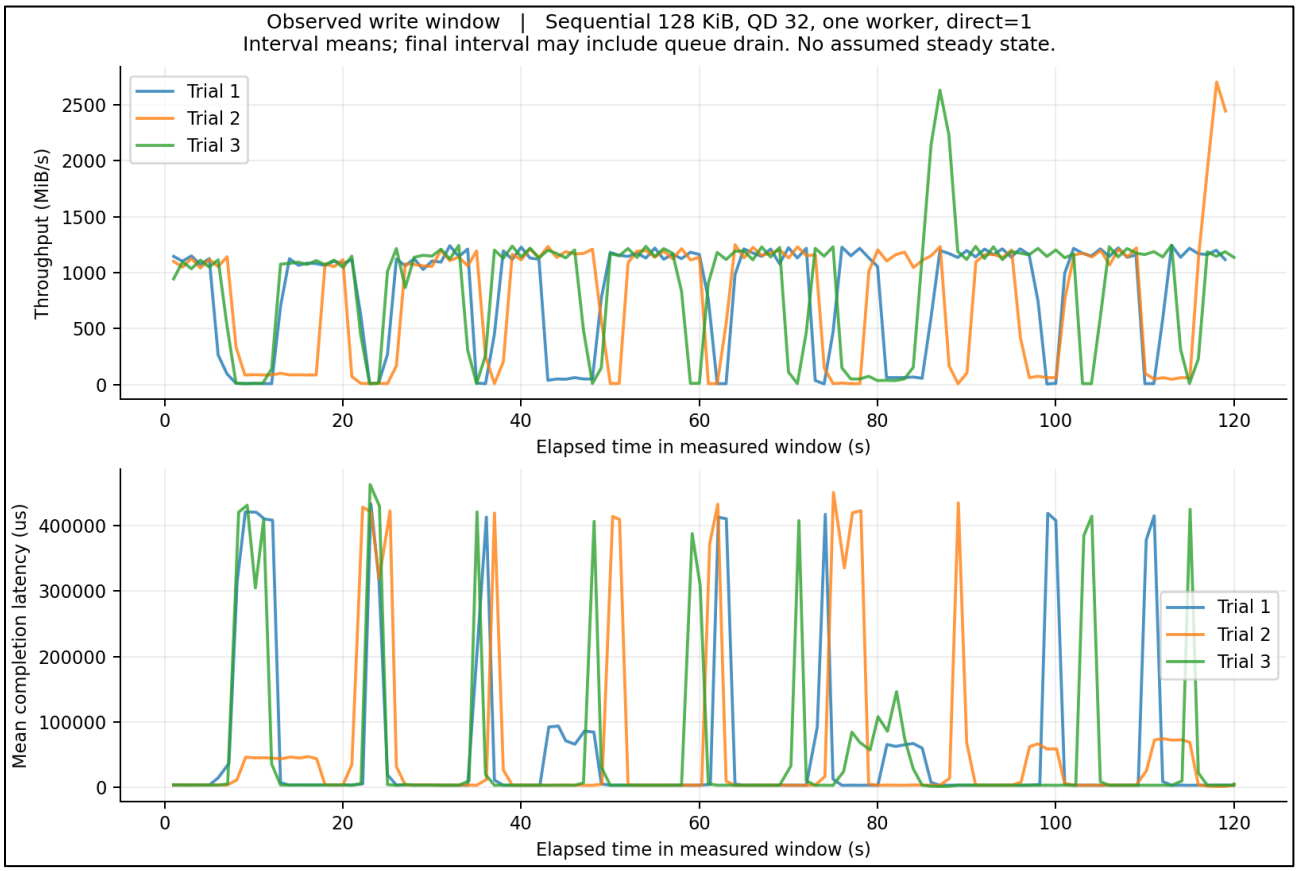


Figure 12. Three 120-second sequential-write windows at 128 KiB and QD 32. One-second interval means show recurring throughput drops and completion-latency spikes.

Trial	First 40 s MiB/s	Last 40 s MiB/s	First 40 s clat ms	Last 40 s clat ms
1	762.0758	868.8379	86.7176	47.2945
2	625.1461	857.7241	67.1701	30.6881
3	818.1303	1004.0766	76.6124	39.4968

All three late windows have higher throughput and lower interval-mean completion latency than their early windows. The recording therefore does not establish a simple burst-then-slower-steady-state transition or an exhausted SLC cache. Instead, recurring low-throughput intervals interrupt periods near the higher sustained rate. The whole-run write mean is 831.9722 ± 45.6632 MiB/s, with mean total latency 4792.3308 ± 261.6226 µs.

Window statistics weight each logged interval by its overlap with the first or last 40 seconds. The latency columns average interval means over time; they are not I/O-count-weighted whole-run latency, so they can substantially exceed the whole-run mean during low-throughput stalls. Queue drain after the configured 120 seconds receives zero weight. Endpoint throughput spikes in the figure should not be interpreted as a stable rate.

Garbage collection, write-cache management, thermal throttling and host scheduling are possible explanations, not established causes. No guest temperature sensors were available. The one pre-run host reading of 40 °C and its reported TemperatureMax field cannot establish temperatures during these pauses. No full-drive preconditioning, physical NAND-write counter or steady-state criterion was measured.

Benchmark Pitfall

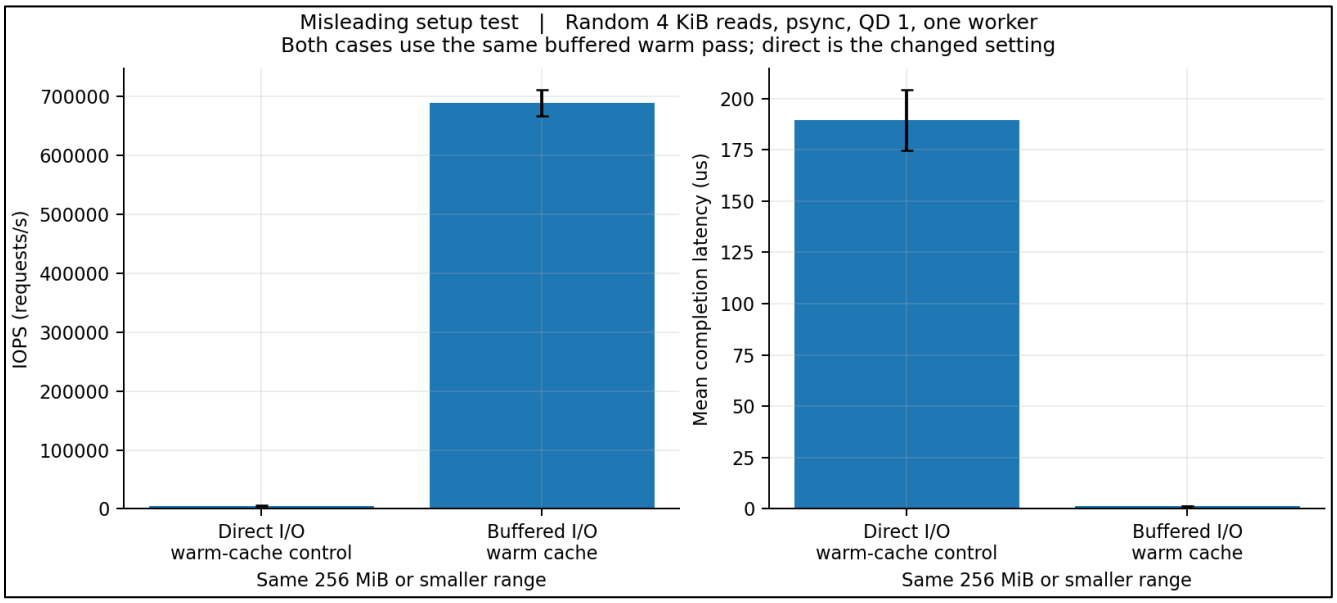


Figure 13. Warmed buffered reads versus a warmed direct-I/O control, using the same 256 MiB range, psync engine, QD 1 and one worker.

Buffered reads reach 689616.8033 ± 22156.7973 IOPS and 1.1973 ± 0.0391 µs mean completion latency. Direct reads reach 5252.2276 ± 393.8445 IOPS and 189.4858 ± 14.8453 µs. The ratio of mean rates is 131.2999×. Calling the buffered rate SSD IOPS would overstate the direct-control mean by 13029.9870%.

Both controls receive a buffered warm pass before timing. Keeping the range, engine and queue settings fixed makes the direct flag the intended changed variable. The large gap demonstrates the misleading claim: a cache-resident read result cannot establish physical SSD capability. Direct I/O bypasses the guest page cache, but does not remove every possible host or device cache and does not by itself guarantee write durability. [1]

Validation and Reproducibility

The supplied archive contains the exact generated fio jobs, raw JSON+, stderr, actual-order journal, source hashes, before/after telemetry, adaptive selections and parsed tables. All 315 successful measured raw files matched their journal SHA-256 hashes. Re-parsing them reproduced 642 measurement rows and all 20,330 numeric fields across 214 summary rows. Every one of the 105 configurations has three successful measured trials. Coverage reports no missing runs and confirms the full default matrix.

One earlier attempt was rejected by the original parser after fio completed successfully: clat histogram N exceeded total_ios by 15 at per-worker QD 16. The compatibility correction retains histogram samples and accepts only a positive ramp-boundary excess bounded by QD − 1 per libaio worker, while requiring histogram counts to equal clat N. The manifest preserves the old fingerprint and changed-source provenance. The failed attempt remains archived; its replacement is used, and the four completed baselines were preserved.

The 512 diagnostic notes comprise 426 ramp-boundary accounting notes, 85 small-sample tail warnings and that one historical execution-failure record. They are not 512 missing or failed retained measurements. The retained jobs report no fio job errors. Across 631 before/after telemetry snapshots, minimum guest MemAvailable is 5,938,812 KiB and recorded swap use is zero; this does not rule out activity between snapshots or paging on the Windows host.

Coverage records 1193.7417 GiB of measured writes, excluding ramps and initialization. This is logical write traffic counted by fio, not physical NAND wear or write amplification. The source and runnable scripts accompany the original project package; this report and the uploaded results archive supply the empirical evidence. The full run can be reproduced with the full profile, the same 16 GiB managed file, seed 7530 and recorded metadata, although host state cannot be recreated exactly.

Conclusion

The objective was to explain how workload and concurrency determine SSD performance rather than report one peak number. In the tested WSL path, larger blocks increase byte throughput while reducing request rate. Read/write mixes perform worse than either pure endpoint, and additional queue depth eventually increases response time without a useful throughput gain. The declared median-based knee candidates are QD 128 for random reads and writes and QD 32 for sequential reads; a sequential-write knee is not established.

The largest observed median small-block rates are 66808.3589 read IOPS and 37558.5177 write IOPS, both at random 4 KiB QD 128. The largest median sequential throughput is 2832.7723 MiB/s for 256 KiB reads at QD 16 and 912.5920 MiB/s for 128 KiB writes at QD 8. These maxima come from the block, queue and parallel families, not a manufacturer specification or a bare-device ceiling. The read maximum does not establish a PCIe limit: negotiated link width/speed and isolated host/device utilization were not recorded.

Tail behavior limits what these rates mean for responsive applications. Sequential-write p99.9 exceeds two seconds at the highest-tested tail depth even though its p50 is 28.5300 ms. Sustained writes show recurring stalls without a consistent early-to-late slowdown. The cache control exaggerates direct-read IOPS by more than two orders of magnitude. Practical settings should therefore balance byte rate, request rate and tail response, and identify the caching and virtualization layers included in the measurement.

Measurement Limits

The results characterize file I/O through WSL2 and its virtual disk, with ext4 in the guest and a Windows storage path below it. The host lists one NVMe SSD, but the VHDX location and exact physical-volume mapping were not recorded. Interface generation, link width, controller utilization, drive encryption/compression settings and host cache policy remain unverified. C: free space was captured before the run, with no matching post-run Windows-volume reading. Linux free space describes the guest filesystem rather than guaranteed physical host capacity. [2]

The operator-notes template was not filled in. AC power, cooling, unrelated background work and environment changes are therefore not established; Balanced power mode is supported only by the earlier host snapshot. Full-drive history, secure erase and device-level steady-state conditioning were not controlled. A full-file initialization establishes written test data, not uniform physical flash state. There are no continuous SSD temperature, power, physical-write or garbage-collection measurements to isolate the source of write stalls.

Three trials and large SDs limit fine comparisons. QD 1 tails often have small populations, while high-QD samples include a bounded number of completions crossing the ramp boundary. Longer tail trials and short queue trials were collected at different times and are kept separate. Queue-depth occupancy is evaluated at fio's completion boundary, not the physical controller; direct-write completion is not equivalent to a per-request persistent-media guarantee.

AI Assistance

ChatGPT generated the benchmark framework and supplemental scripts, helped correct fio compatibility and histogram accounting, and assisted with analysis and report wording. Measurements come from the laptop's WSL run. Reported values use the supplied CSVs and raw JSON+ rather than plot estimates. All retained raw hashes and summary numeric fields were checked; unrecorded operating conditions are identified as limits rather than supplied as assumptions.

References

[1] fio project. Flexible I/O Tester documentation. I/O engines, direct I/O, queue depth, ramp time, latency statistics and JSON+ output. https://fio.readthedocs.io/en/latest/fio_doc.html. Accessed October 24, 2026. The experiment used fio 3.28; saved job files and raw output establish its actual options and measurements.

[2] Microsoft Learn. How to manage WSL disk space. WSL2 virtual disks, ext4.vhdx and guest-versus-host capacity. https://learn.microsoft.com/en-us/windows/wsl/disk-space. Accessed October 24, 2026.